Upgrading Urban Air Quality Models: A Two-Layer Formulation for Boundary Layer Entrainment and Morning Fumigation

1. The Physical and Computational Imperative for Vertical Structure

Accurate prediction of urban air quality, particularly during acute high-pollution transport episodes, depends heavily on the robust simulation of atmospheric boundary layer dynamics. The boundary layer acts as the fundamental mediator between surface emissions, atmospheric chemistry, and regional transport. In many simplified operational air quality models, vertical structure is diagnosed strictly to define a volumetric capacity for a single well-mixed layer, while chemical solving and transport are collapsed into this single dimension. A common computational shortcut in such systems is the use of a "threshold gate" on the inversion lid: transported plumes arriving above the nocturnal inversion contribute nothing to surface concentrations until the mixing height abruptly eclipses the plume height, at which point the pollutant mass is instantly dumped into the surface layer.

While computationally frugal, this threshold gate represents a profound physical fallacy. Atmospheric mixing is not a binary switch but a continuous, rate-controlled process driven by turbulent kinetic energy, thermal buoyancy, and wind shear. The boundary layer entrains air continuously as it grows, diluting surface emissions with clean air or, in the case of transported smoke, fumigating the surface with aloft pollutants. By utilizing a threshold gate, a model introduces artificial numerical shocks, misses the gradual dilution effect of an expanding boundary volume, and completely misrepresents the timing and amplitude of the morning fumigation peak—a critical period for public health exposure.

This report evaluates the minimum defensible upgrade from a single-layer threshold model to a multi-layer slab model. It comprehensively details published two-layer and three-layer formulations, the parametrization and timescales of morning fumigation, observational evidence of elevated smoke layers over Delhi from campaigns such as APHH-India, and a rigorous assessment of whether this upgrade measurably improves surface particulate matter ($\mathrm{PM}_{2.5}$) predictions. Finally, it provides a specific, highly tractable two-layer formulation with governing equations and parameter ranges designed for rapid implementation, alongside an honest assessment of whether the upgrade justifies the added computational machinery.

2. Conceptual Framework: Box Models Versus Multi-Layer Slab Models

2.1 The Legacy and Limitations of the Single-Layer Box Model

The simplest screening tool for estimating average urban pollutant concentration is the steady-state box model, which treats the air above a city as a single well-mixed control volume[1]. The fundamental premise of the box model is the conservation of mass within a defined spatial domain. The concentration C within this volume is dictated by the mass balance between area emissions q, the along-wind length of the city L, the ventilating wind speed u, the mixing height H, and the upwind background concentration C_b, yielding the fundamental relationship $C = qL/(uH) + C_b$ [1].

While mathematically elegant and useful for rapid sensitivity checks regarding stagnation risks, the single-layer box model assumes instantaneous and complete vertical mixing up to H [1]. It fails entirely to account for the diurnal evolution of the atmospheric boundary layer, which transitions from a deep convective mixed layer during the day to a shallow, stable nocturnal boundary layer capped by a residual layer at night[4]. When the single-layer model relies on a static or step-wise prescribed boundary layer height, it fundamentally cannot conserve mass correctly across the diurnal transition. Most importantly, it completely loses track of the fate of pollutants that become trapped aloft when the nocturnal inversion forms, treating the atmosphere above the nocturnal boundary layer as a void.

2.2 The Evolution to Two-Layer and Three-Layer Slab Models

To bridge the gap between rudimentary box models and computationally expensive three-dimensional Eulerian Chemical Transport Models, the atmospheric science community developed multi-layer "slab" models. These models, which date back to foundational work by Tennekes (1973) and were later refined into contemporary frameworks like the Chemistry Land-surface Atmosphere Soil Slab (CLASS) model by Vilà-Guerau de Arellano et al., represent the boundary layer using vertically integrated differential equations[7].

The CLASS model and its derivatives are conceptual bulk models that rely on mixed-layer equations[7]. They assume that turbulence inside the convective boundary layer is sufficiently strong to homogenize potential temperature (θ), specific humidity (q), and pollutant concentrations (C)[8]. Crucially, they explicitly model the interface between the surface mixed layer and the free troposphere (or residual layer) as an infinitesimally thin inversion zone characterized by a "jump" in scalar values[8].

In a standard two-layer formulation, the lower layer is the active mixed layer, and the upper layer is the residual layer. The residual layer acts as a temporary chemical reservoir. At sunset, when surface heating ceases and convective turbulence decays, the upper portion of the daytime mixed layer is decoupled from the surface, becoming the residual layer, while a shallow stable layer forms at the ground[5]. Pollutants emitted during the night accumulate heavily in the shallow surface layer, while pollutants transported from distant sources advect into the residual

layer[5].
Advanced three-layer models further subdivide the atmospheric column to capture steep near-surface gradients. In a three-layer framework, the system explicitly resolves an atmospheric surface layer (the lowest 10% of the boundary layer where shear dominates and scalar profiles are logarithmic), the convective mixed layer, and the entrainment zone or residual layer aloft[7]. While three-layer models are superior for simulating the deposition fluxes of reactive gases like ozone, a two-layer formulation (mixed layer plus residual layer) is generally considered the optimal balance of complexity and performance for tracking the mass balance of inert or long-lived aerosols like transported $\mathrm{PM}_{2.5}$ [12].

Model Type	Vertical Resolution	Treatment of Aloft Transport	Primary Use Case
Single-Layer Box	One well-mixed volume up to H.	Threshold gate or entirely ignored.	First-order screening; steady-state urban averages.
Two-Layer Slab	Mixed Layer + Residual Layer/Free Troposphere.	Stored in the upper layer; mixed downward via entrainment velocity.	Diurnal air quality forecasting; regional transport episodes.
Three-Layer Slab	Surface Layer + Mixed Layer + Residual Layer.	Same as two-layer, but resolves steep near-surface gradients.	Deposition modelling; biosphere-atmosphere chemical exchange.

3. Parameterization of Entrainment and Morning Fumigation

3.1 The Physics of the Entrainment Velocity Parameterization

The exchange of mass between the residual layer and the mixed layer is dictated strictly by the entrainment velocity, w_e. Entrainment is the process by which the growing boundary layer incorporates fluid from the quiescent layer above it[7]. In operational systems, parameterizing this velocity correctly is the cornerstone of modeling morning fumigation.
The standard parameterization for entrainment relies on the turbulent kinetic energy budget at

the inversion interface. A widely adopted and highly robust formulation is the slab model of Batchvarova and Gryning (1991, 1994), which calculates the boundary layer growth rate based on both convective buoyancy and mechanical wind shear[7]. In this framework, the entrainment velocity is fundamentally linked to the surface sensible heat flux ($\overline{w'\theta'}_s$) and the strength of the temperature inversion capping the boundary layer ($\Delta\theta$).
A simplified form of the Batchvarova and Gryning entrainment parametrization for a purely convective boundary layer is expressed as:

$$w_e = \frac{\beta \overline{w'\theta'}_s}{\Delta\theta}$$

where β is the entrainment flux ratio, empirically found to be approximately 0.2[8]. This signifies that the downward heat flux at the inversion is roughly 20% of the upward surface heat flux[8]. However, in operational regulatory models such as CALPUFF and various Lagrangian Particle Dispersion Models, mechanical turbulence is also significant, and the full Batchvarova and Gryning closure is utilized[16]. The governing equation for the boundary layer depth h is given by:

$$\frac{dh}{dt} = \frac{(1+2A)\overline{w'\theta'}_s + 2B\frac{u_*^3 T}{gh}}{\Delta\theta + C\frac{\gamma u_*^2 T}{g}}$$

where h is the boundary layer depth, u_* is the friction velocity, T is the surface temperature, g is the acceleration due to gravity, and γ is the lapse rate of potential temperature in the overlying layer[16].

Parameter	**Symbol**	**Typical Operational Value**	**Physical Meaning**
Convective Closure	A	0.20	Ratio of entrainment heat flux to surface heat flux.

Mechanical Closure	B	2.50	Efficiency of shear-driven turbulence in boundary layer growth.
Stability Parameter	C	8.00	Damping effect of stable stratification on shear mixing.

The entrainment velocity is then formally defined as $w_e = \frac{dh}{dt} - w_s$, where w_s is the large-scale subsidence velocity[8]. In many simplified urban models, subsidence is neglected ($w_s = 0$), making the entrainment velocity mathematically equivalent to the boundary layer growth rate during the daytime expansion phase[18].

3.2 The Timescale and Dynamics of Morning Fumigation

Fumigation is the downward mixing of elevated pollutant plumes to the ground as the convective boundary layer grows and penetrates the residual layer[19]. In operational systems, this is not modeled as an instantaneous event, but rather as a highly transient, continuous rate-controlled process governed by w_e .

In subtropical urban environments like Delhi, the timescale over which morning fumigation occurs is remarkably consistent, though highly dependent on season. Following sunrise, surface heating requires a latency period of approximately 1 to 2 hours to erode the strong, shallow nocturnal surface inversion[21]. Once the surface inversion is broken, the convective boundary layer grows rapidly into the neutrally stratified residual layer left over from the previous day[6].

Observational studies utilizing micropulse lidar and ceilometers indicate that this rapid growth phase typically begins around 07:00 to 08:00 Local Time[21]. During this phase, boundary layer growth rates during clear-sky conditions can exceed 380 meters per hour[22]. Consequently, the entire residual layer, which often extends to altitudes of 1,000 to 1,500 meters, is engulfed within a highly active 2- to 3-hour window. The fumigation event flushes the aloft pollutants into the surface layer, leading to a sharp, sustained spike in ground-level concentrations between 08:00 and 10:00 Local Time, before daytime dilution via continued boundary layer deepening takes over[21].

4. Observational Evidence for Delhi: Elevated Smoke and Downward Mixing

Delhi suffers from extreme $\mathrm{PM}_{2.5}$ concentrations, particularly during the post-monsoon and winter seasons. The atmospheric pollution profile is driven by a complex interplay of local urban emissions and the regional transport of biomass burning smoke (crop stubble fires) from the upwind states of Punjab and Haryana[21]. The vertical stratification of this smoke, and its subsequent entrainment, is the central mechanism dictating the city's diurnal pollution extremes.

4.1 Lidar, Ceilometer, and Aircraft Profiling Campaigns

Extensive observational evidence confirms the presence of massive elevated smoke layers over the Indo-Gangetic Plain that remain entirely decoupled from the surface during the night. The NERC-funded Atmospheric Pollution and Human Health in an Indian Megacity (APHH-India) campaign deployed an unprecedented suite of advanced instrumentation, including ceilometers, Doppler lidars, and the BAe-146 research aircraft, to capture the vertical distribution of aerosols over Delhi[25].

Aircraft profiling during the APHH-India campaign revealed highly complex vertical structures. Significant concentrations of secondary organic aerosols, black carbon, and specific biogenic markers were located high in the residual layer[25]. High-resolution liquid chromatography coupled to mass spectrometry identified significant formations of isoprene organosulfates and nitrooxy-organosulfates aloft[25]. These findings proved that transported agricultural plumes undergo extensive chemical processing in the residual layer overnight, entirely isolated from the surface chemistry.

Ground-based remote sensing using mini micropulse lidar and HALO Doppler lidar in similar Indian coastal and inland environments routinely identifies double mixed-layer characteristics during the morning transition: a growing internal convective boundary layer at the surface, and a distinct, heavily polluted residual layer aloft[22]. Lidar depolarization ratios have been utilized to differentiate between fresh smoke, aged smoke, and coarse dust, confirming that advected biomass burning plumes frequently reside in the residual layer overnight at altitudes of 500 to 1,500 meters[22]. Because the nocturnal boundary layer in Delhi during winter can be exceptionally shallow, sometimes collapsing to less than 50 meters deep, these massive smoke plumes pass entirely overhead during the night without impacting surface monitoring stations[12].

4.2 The Surface Signature of Fumigation

The descent of these elevated layers leaves a distinct, unavoidable temporal fingerprint on ground-level monitors. Diurnal variations of $\mathrm{PM}_{2.5}$ and Black Carbon in Delhi consistently exhibit a bimodal "W" shape[24]. While the evening peak is primarily driven by the collapse of the boundary layer and the trapping of local rush-hour emissions in a shrinking volume, the morning peak is heavily influenced by the fumigation of the residual layer[22].

Surface observations show a sharp peak in $\mathrm{PM}_{2.5}$ and Black Carbon mass concentrations between 07:00 and 10:00 Local Time[21]. This peak cannot be explained by morning rush-hour traffic alone. Budget analyses and source apportionment confirm that the rapid increase in surface mass is driven largely by the entrainment of the aloft smoke plume as the boundary layer grows[23]. The fumigation effect is so pronounced that the downward mixing of residual layer aerosols can contribute the overwhelming majority of the surface-level $\mathrm{PM}_{2.5}$ during the morning transition[12]. Such significant peaks in mornings are the direct result of breaking off the nocturnal boundary layer and the onset of convective turbulence[27].

Diurnal Phase	Local Time (Delhi)	Boundary Layer Dynamics	Dominant PM2.5 Influence
Nocturnal	22:00 – 06:00	Stable, shallow surface layer; decoupled residual layer aloft.	Accumulation of local emissions at surface; smoke transport aloft.
Transition/Fumigation	07:00 – 10:00	Rapid convective growth; erosion of inversion; high w_e.	Aloft smoke mixed downward; sharp spike in surface concentrations.
Midday Dilution	11:00 – 16:00	Deep, fully mixed convective boundary layer (1.5 km).	Maximum volumetric dilution; lowest surface concentrations.
Evening Collapse	17:00 – 21:00	Decay of turbulence; formation of new nocturnal inversion.	Trapping of evening rush-hour emissions; secondary peak.

5. Does a Two-Layer Scheme Measurably Improve Predictions?

When evaluating the architectural upgrade of an operational model, the central question is whether the gain is purely conceptual or if it translates to measurable predictive improvement during critical transport episodes. The published literature demonstrates unequivocally that resolving the residual layer is mathematically essential for accurate diurnal forecasting.

5.1 Correcting Diurnal Amplitudes and Peak Timing

Single-layer models frequently fail to replicate the diurnal concentration profile, leading to severe mischaracterizations of public exposure. A comprehensive study utilizing the

GEOS-Chem high-performance (GCHP) chemical transport model demonstrated that baseline single-layer simulations significantly overestimate nighttime $\mathrm{PM}_{2.5}$ accumulation[23]. Because a single-layer model compresses all regional emissions and transported mass into an unrealistically small surface volume at night, while simultaneously failing to store advected smoke aloft, it generates excessive diurnal amplitudes and premature morning concentration peaks[23].

Budget analyses in advanced modelling frameworks indicate that the morning peak of $\mathrm{PM}_{2.5}$ is fundamentally driven by changes in the aerosol subgrid vertical gradient coupled with fumigation after sunrise[23]. By introducing a multi-layer approach or explicit boundary layer mixing schemes, models are able to hold the transported smoke aloft in a simulated residual layer. This physical separation prevents the artificial nighttime spike at the surface and accurately releases the pollutant mass during the 07:00–10:00 Local Time window. The improvement in the correlation coefficient and the reduction in Mean Fractional Bias for diurnal profiling are statistically significant when the residual layer reservoir is explicitly resolved[23]. Furthermore, decreased boundary layer mixing is commonly discussed as a contributor for the $\mathrm{PM}_{2.5}$ increase during the evening, highlighting the necessity of capturing the exact temporal decay of turbulence[23].

5.2 The Physical Fallacy of the Threshold Gate

The "threshold gate" utilized in simplified computational systems operates on a binary conditional logic, typically triggering a mass release when the mixing height surpasses the plume height. This creates two severe physical fallacies that degrade predictive accuracy:

1. **Numerical Shocks:** When the boundary layer height crosses the gate threshold, the entire mass of the plume is instantaneously mixed into the surface volume. This causes a discontinuous, unphysical spike in surface $\mathrm{PM}_{2.5}$ that triggers false alarms in Air Quality Index reporting.
2. **Missing Dilution:** A growing boundary layer does two things simultaneously: it entrains polluted air from the aloft plume, but it also increases the total volume of the surface layer. Entrainment is a continuous rate process[7]. A threshold gate completely ignores the continuous volumetric dilution that occurs before and after the gate is triggered.

A two-layer slab scheme mathematically links the mass transfer directly to the continuous boundary layer growth rate (dh/dt). This ensures that the fumigation is properly rate-limited and that the simultaneous effect of volume expansion is natively handled[7]. For a regulatory or forecasting model predicting acute morning exposure, the difference between an instantaneous gate-dump and a rate-controlled 2-hour fumigation profile is the difference between a numerical artifact and a physically sound prediction.

6. Recommended Two-Layer Formulation and Implementation Guide

The transition from a threshold gate to a two-layer slab model is the minimum defensible upgrade for any system attempting to model regional transport. It introduces no requirement for 3D advection solvers, complex vertical diffusivity matrices, or heavy computational overhead. Instead, it relies on solving ordinary differential equations for the mass budgets of two interacting control volumes: the Mixed Layer and the Residual Layer.

6.1 The Governing Equations

Let C_m be the pollutant concentration in the mixed layer and C_r be the concentration in the residual layer. Let h be the time-dependent depth of the mixed layer. The total depth of the domain (Mixed Layer + Residual Layer) can be fixed at a height H_{top} (e.g., 2,000 meters, representing the maximum daytime boundary layer height).

The Mixed Layer Mass Budget:

The total mass in a unit column of the mixed layer is the product of the depth and the concentration, $h \cdot C_m$. The time rate of change of this mass is driven by surface emissions (E), dry deposition ($v_d C_m$), chemical production or loss (S_m), and the entrainment of mass from the residual layer ($w_e C_r$):

$$\frac{d(hC_m)}{dt} = E - v_d C_m + S_m h + w_e C_r$$

Applying the product rule to the left side ($\frac{d(hC_m)}{dt} = h\frac{dC_m}{dt} + C_m \frac{dh}{dt}$) and recognizing that the boundary layer growth rate $\frac{dh}{dt} \approx w_e$ (assuming large-scale subsidence $w_s = 0$ for simplicity in a basic box model), we obtain the prognostic equation for the mixed layer concentration[7]:

$$h\frac{dC_m}{dt} = E - v_d C_m + S_m h + w_e(C_r - C_m)$$

The elegance of this formulation lies in the entrainment term, $w_e(C_r - C_m)$. This term natively handles both fumigation (when the residual layer is more polluted, $C_r > C_m$) and clean-air dilution (when the residual layer is pristine, $C_r < C_m$). When the boundary layer is not growing ($w_e = 0$), this term vanishes entirely, and the layer operates as a standard

single-layer box model.
The Residual Layer Mass Budget:

The residual layer occupies the volume between the top of the mixed layer h and the domain top H_{top}. Its depth is $h_r = H_{top} - h$. The mass balance for the residual layer is driven by the horizontal advection of transported smoke (A_r), chemical transformations (S_r), and the loss of mass to the mixed layer due to entrainment.
Because the boundary layer is growing into the residual layer, the physical volume of the residual layer shrinks. However, the *concentration* of the residual layer does not change due to this volume loss; it simply loses mass at the lower boundary. Therefore, the prognostic equation for the residual layer concentration is independent of w_e:

$$\frac{dC_r}{dt} = A_r + S_r$$

where A_r represents the source term for the advected stubble smoke arriving aloft over the city.
The Inversion Jump: Advanced slab models, such as CLASS, often track the scalar jump at the inversion, defined as $\Delta C = C_r - C_m$. Taking the derivative of this provides a coupled differential equation that is often more numerically stable in rigid computational solvers[8]:

$$\frac{d\Delta C}{dt} = \frac{dC_r}{dt} - \frac{dC_m}{dt}$$

$$\frac{d\Delta C}{dt} = A_r + S_r - \frac{1}{h}\left[E - v_d C_m + S_m h + w_e \Delta C\right]$$

6.2 Parameterizing the Entrainment Velocity (w_e)

To mathematically close the system, the entrainment velocity w_e must be supplied. If the existing system already diagnoses the mixing depth $h(t)$ dynamically from meteorological inputs, w_e can simply be calculated as the temporal derivative of the mixing height during growth periods:

$$w_e = \max\left(\frac{dh}{dt}, 0\right)$$

A crucial physical constraint must be enforced: entrainment only occurs when the boundary layer is growing[5]. When the boundary layer collapses at sunset, $w_e = 0$. The mass that was in the mixed layer above the new, shallow nocturnal boundary layer height is instantaneously transferred to the residual layer[5]. If the system diagnoses static stability but does not provide a smooth $h(t)$, the simplified Batchvarova and Gryning formulation should be implemented: $w_e = 0.2\overline{w'\theta'_s}/\Delta\theta$ [15].

6.3 Implementation Mechanics and Parameter Ranges

This two-layer upgrade can be implemented in fewer than 200 lines of code. The execution sequence at each time step Δt follows a strict logical flow:

1. **Meteorological Update:** Update emissions E, heat fluxes, and diagnose the current mixing height h and entrainment velocity w_e.
2. **Evening Transition Logic:** If the sun sets and h drops from its daytime maximum (h_{day}) to its nocturnal minimum (h_{night}), update the residual layer concentration by volume-weighting the old residual layer and the abandoned portion of the daytime mixed layer:

$$C_{r,new} = \frac{C_r(H_{top} - h_{day}) + C_m(h_{day} - h_{night})}{H_{top} - h_{night}}$$

3. **Numerical Integration:** Use an explicit Euler or second-order Runge-Kutta scheme to integrate the mass budget differential equations. Because the entrainment term w_e/h can become large when h is very small (e.g., immediately at sunrise), use a time step Δt of 1 to 5 minutes, or enforce a minimum boundary layer height (e.g., $h_{min} = 50$ meters) to ensure strict numerical stability.

To assist with initialization and bounding, typical parameter values for a subtropical megacity during the post-monsoon transport season are provided below based on observational literature.

Parameter	Symbol	Typical Value (Delhi Post-Monsoon)
Nocturnal Boundary Layer Height	h_{night}	50 – 150 meters
Daytime Maximum Mixing Height	$h_{day,max}$	1,000 – 1,500 meters
Morning Entrainment Velocity	w_e	0.05 – 0.15 m/s (180 – 540 m/hr)
Domain Top Height	H_{top}	2,000 meters
Morning Fumigation Window	t_{fum}	07:00 – 10:00 Local Time

7. Final Assessment: Justification of the Upgrade

The inclusion of the two-layer formulation described above is entirely defensible and heavily justified. The current threshold gate approximation is computationally convenient but physically indefensible for regions like the Indo-Gangetic Plain, where acute morning exposure is driven precisely by the rate of boundary layer entrainment of aloft smoke[21].

The added computational machinery is minimal. It requires tracking exactly one new state variable (the residual layer concentration C_r) and replacing the binary threshold logic with standard ordinary differential equations for mass balance that utilize the already-diagnosed boundary layer growth rate[7]. This upgrade will seamlessly and naturally handle both the continuous dilution of the surface volume during the day and the exact timing and magnitude of morning fumigation[7].

Crucially, implementing this two-layer scheme will smooth out the artificial concentration spikes caused by the threshold gate, correct the severe diurnal amplitude errors inherent to single-layer models, and align the system with the foundational mixed-layer physics utilized in professional assessment tools like the CLASS model[8]. Documenting the lid gate as an "acceptable approximation" is no longer scientifically valid when the dominant contributor to surface $\mathrm{PM}_{2.5}$ during transport episodes is the transient entrainment of the residual layer. The two-layer slab model represents the necessary and highly efficient evolution of the system.

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